whereas,

$$\begin{aligned} e^{hA}e^{hB} &= (\mathrm{Id} + hA + \frac{h^2A^2}{2} + \frac{h^3A^3}{6} + \mathcal{O}(h^4)) \\ &\quad \times(\mathrm{Id} + hB + \frac{h^2B^2}{2} + \frac{h^3B^3}{6} + \mathcal{O}(h^4)) \\ &= \mathrm{Id} + h(A+B) + \frac{h^2}{2}(2AB + A^2 + B^2) \\ &\quad + \frac{h^3}{6}(A^3 + B^3 + 3AB^2 + 3A^2B) + \mathcal{O}(h^4). \end{aligned}$$

So the difference is

$$\begin{aligned} e^{hA}e^{hB} - e^{h(A+B)} &= \frac{h^2}{2}(AB - BA) \\ &\quad + \frac{h^3}{6}(2AB^2 + 2A^2B - BA^2 - BAB - B^2A - ABA) + \mathcal{O}(h^4) \\ &= \frac{h^2}{2}[A,B] + \mathcal{O}(h^3), \end{aligned}$$

where $[A,B] = AB - BA$ is the commutator of A and B.

For a splitting method, we have, replacing A by $\mathcal{L}_{H_1}$ and B by $\mathcal{L}_{H_2}$,

$$e^{h\mathcal{L}_{H_1}}e^{h\mathcal{L}_{H_2}} - e^{h\mathcal{L}_H} = \frac{h^2}{2}[\mathcal{L}_{H_1}, \mathcal{L}_{H_2}] + \mathcal{O}(h^3).$$

Observe that, for any real-valued functions f and H of phase space,

$$\mathcal{L}_H f = \nabla f^T \boldsymbol{J} \nabla H = \{f, H\}.$$

Hence

$$\mathcal{L}_{H_1}\mathcal{L}_{H_2} f = \{\{f, H_2\}, H_1\},$$

and, using this and skew-symmetry of the Poisson bracket,

$$[\mathcal{L}_{H_1}, \mathcal{L}_{H_2}]f = \{\{f, H_2\}, H_1\} - \{\{f, H_1\}, H_2\} = \{\{f, H_2\}, H_1\} + \{\{H_1, f\}, H_2\}.$$

By the Jacobi identity,

$$\{\{f, H_2\}, H_1\} + \{\{H_2, H_1\}, f\} + \{\{H_1, f\}, H_2\} = 0.$$